Controlled re-test of the NIST cyclotomic / point-height note

Date. 2026-10-11. **What it tests.** The comparative and "nothing found" claims of `RESEARCH_NIST_CYCLOTOMIC_POINT_HEIGHTS.md` (merged in #1655). Each claim gets a null population, a positive control and a frozen decision rule, following `.claude/skills/controlled-experiment/SKILL.md`, which this change introduces. **Status.** Measured. No attack is claimed and no ECDLP cost changes.

Verdicts in one table

original claim (note §)	**class**	**verdict (pre-registered rules; v2 unless v3 is named)**
P-256 $n - 1$ is "the outlier", "a 1-in-30 accident" (§0, §2.2)	comparative	**Not established.** It ranks 1992/2000 on M1 (p = 0.009) and 1991/2000 on M2 (p = 0.010) against random Hasse-interval primes. α = 0.05/18 = 0.0028, so it is **within the control distribution**. It is more extreme than 99.6% of controls before correction. The "1-in-30" Dickman figure assumed the wrong null and is withdrawn. **v3 (ECM-completed factoring):** same verdict, rank 796/800, p = 0.0125 > 0.05/9. ECM changed M1 for 0 of 2400 control values, so the rho instrument was already complete below 2^32.
other $n \pm 1$ and twist orders unremarkable (§2.2, §2.3)	comparative	**Supported.** All 8 are within the control distribution (p ≥ 0.20 on both metrics).
the canonical lift has no small points (§3.1)	"nothing found"	**Supported only as bounded.** The census found 90/90 planted points (integer, $d = 2$, $d = 3$), found nothing on 30 random-b curves, and found nothing on the NIST b. Licensed wording: "no point with integer x, −10^6 ≤ x ≤ 10^6, or $x = u/d^2$ with $d \le 40$ and −10^5 ≤ u ≤ 10^5". "No small point" is not licensed.
lifted generator has $\hat{h} \approx \log p$, nowhere near small (§3.2)	comparative	**Supported as "typical".** $\hat{h}(\tilde{G})/\ln p$ is within the random-point distribution on all three curves (p ≥ 0.37). The generator is an ordinary lifted point, neither large nor small.
two-point lift: independent and orthogonal, cos < 0.002 (§3.3)	comparative	**Independence supported. Orthogonality: typical, measured in v3.** v2's instrument read 1.0011–1.0025 on exactly dependent pairs, so "zero to three decimals" was below its resolution and is withdrawn. **v3** uses rational-model heights, and both accuracy gates passed: dependent pairs are exact to about 1e-16, and the parallelogram defect is at most 2.7e-3 nats. With it, the absolute cosine of ($\tilde{G}$, $\tilde{R}$) is 5.9e-4 / 0 / 4.4e-4 (P-192 / P-224 / P-256), within the random-pair distribution (p ≥ 0.41). For 59–66% of random pairs the canonical pairing is exactly 0; the rest are at most 0.013.

Exact computations in the note (factorisations, certified $Z[t]$ identities, embedding degrees, Cheon tables) are not statistical claims and were not re-tested.

Protocol history (read this first)

1. **`PROTOCOL.md` (v1) was frozen before any data**, and committed in `b8280b445` with sha256 `2c3c8df5…6383`.
2. A small-N dry run on scratch output showed that v1 **could not reject in any comparative family**. Its add-one empirical tail has a minimum two-sided p of `2/(N + 1)`, which is above α/m for E1 (0.0050 vs 0.0028), E3 (0.0100 vs 0.0083) and E4 (0.0198 vs 0.0083). That is a design error.
3. **`AMENDMENT_1.md` (v2) was frozen before any protocol-N run** (commit `49a5e20f7`, sha256 `cbbb4e72…397a`). It raises N only, to 2000 / 1000 / 500 by the rule `N + 1 ≥ 4m/α`, and discloses what the dry run showed, including P-256 `n − 1` lying beyond the dry-run controls. Discount v2 accordingly. The skill now carries this power check.
4. Both versions ran. v1 is reported as frozen. Every v1 verdict is “within the control distribution”, as the power failure guaranteed in advance. v1’s control samples are exact prefixes of v2’s (verified row by row for all 9 curve × experiment arms).

Runs

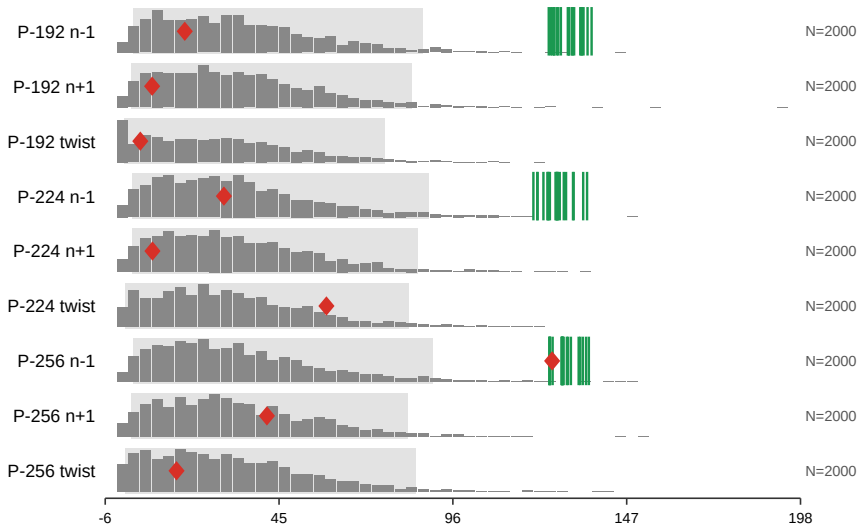
	v1	**v2**
command	`cargo run --release --example nist_cyclotomic_controls -- --protocol v1`	`… -- --protocol v2`
source	working tree committed as `4f068fd76`; later edits only touch the figure layout in `controls_svg`	same binary
host	4-vCPU Intel Xeon @ 2.80 GHz, 15 GiB, Claude Code cloud container, no isolation (wall time is not a result)	same
toolchain	rustc 1.98.1	same
seeds	`20261011 + offset` per experiment and curve	same
N (E1 / E2 / E3 / E4)	400 / 10 per arm / 200 / 100	2000 / not re-run / 1000 / 500
positive controls	E1: 20 planted-smooth per curve. E2: 30 planted per curve. E3: 20 small-x per curve. E4: 2 dependent pairs per curve	same, except E2
completed	all rows; wall 2804 s	all rows; wall 8966 s
records	`rows_v1.jsonl`, `RESULTS_v1.md`, `results_v1.json`	`rows_v2.jsonl`, `RESULTS_v2.md`, `results_v2.json`

All numbers in the tables are measured on the recorded rows. p-values and ranks are derived from them by the frozen rule (`rank_p` in the example). No run failed or timed out.

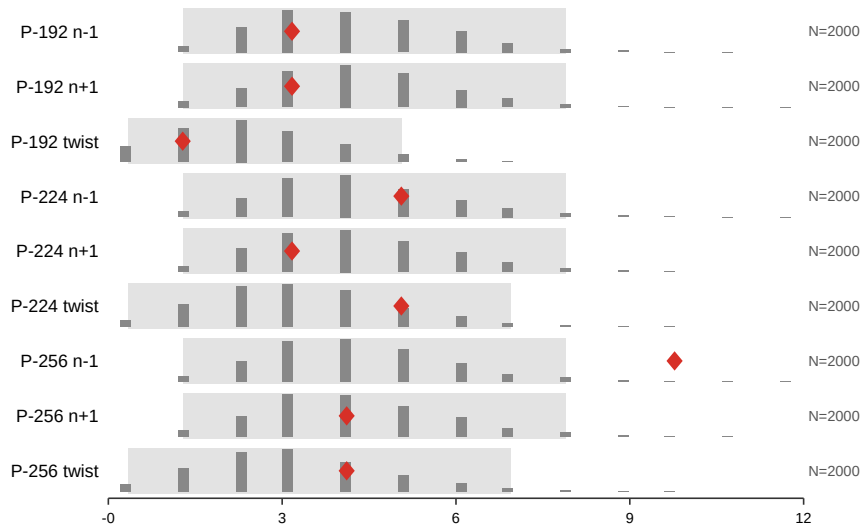
Controlled re-test (v2): control distributions (grey), target (red ◆), positive controls (green |)

Recorded values from rows_v2.jsonl. Band = control 2.5–97.5%. N per row is printed at the right. Seed base 20261011.

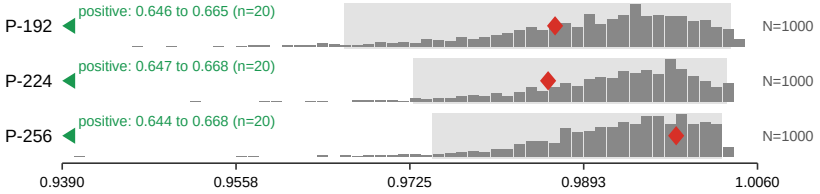
E1 · M1 = log2 of the 2^32-smooth part found (bits)



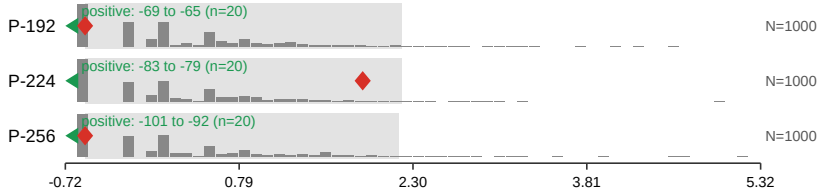
E1 · M2 = # distinct primes below 2^32



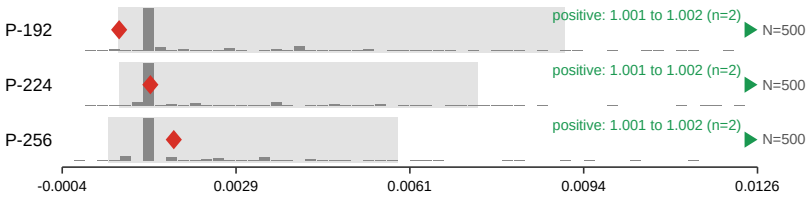
E3 · ĥ/ln p of one-point lift



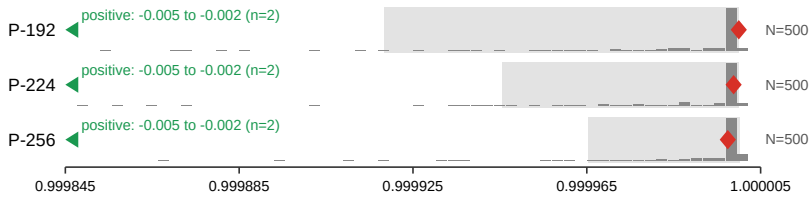
E3 · h_x − ĥ (nats)



E4 · |cos ∠| of two-point lift



E4 · det/(ĥĥ) of two-point lift



Control distributions, targets and positive controls (v2)

Figure (v2). Grey histograms are the control arm per row, with the 2.5–97.5% band shaded. Red diamonds are the NIST targets. Green ticks or edge markers are the positive controls. N is printed per row. Recorded values only. `controls_v1.svg` *is the same layout for v1.*

Observations beyond the protocol (post hoc, labelled)

These were noticed after the data. They are not pre-registered verdicts.

- **The E4 instrument has a floor.** `|cos|` piles up at 0.00128–0.00130 on all three curves (31–43 of 100 v1 pairs per curve hit exactly that value), although $\hat{h}$ differs by a third between curves. The exactly dependent pairs read `|cos|` = 1.0011–1.0025 and det/($\hat{h}\hat{h}$) = −0.002 to −0.005.

  Cause, from reading the code: `height_sequence` scales the curve to an integral model with `u = lcm(all denominators)`, including the point's own (`x.d = d²`, `y.d = d³`). It measures the naive height on that model, so h_x is inflated by about `2 log u` and 4^5 doublings leave about 0.3 nats. On E1 the integer points have `u = 1`, so E3 is unaffected. Amendment 2 (v3, below) measures the reduced rational `x` instead, and re-tests E4 as E4r.

- **`h_x − ĥ` is discrete.** In E3 it takes a few values (−0.5493 = −½ ln 3, −0.2027, 0.1438, …), consistent with local correction terms at 2 and 3. Ties are handled conservatively by the rank rule. P-256's generator sits exactly at the most common value.

- **Small naive height is not small canonical height.** The E3 positive control (points with `x < 2^32`) has `h_x/ln p` of 0.09–0.17 but `ĥ/ln p` of 0.64–0.67, because the lift's `B ≈ y²` is about `p²`. Note §3.4 equated the two and is corrected.

v3: Amendment 2 (E1b, E4r)

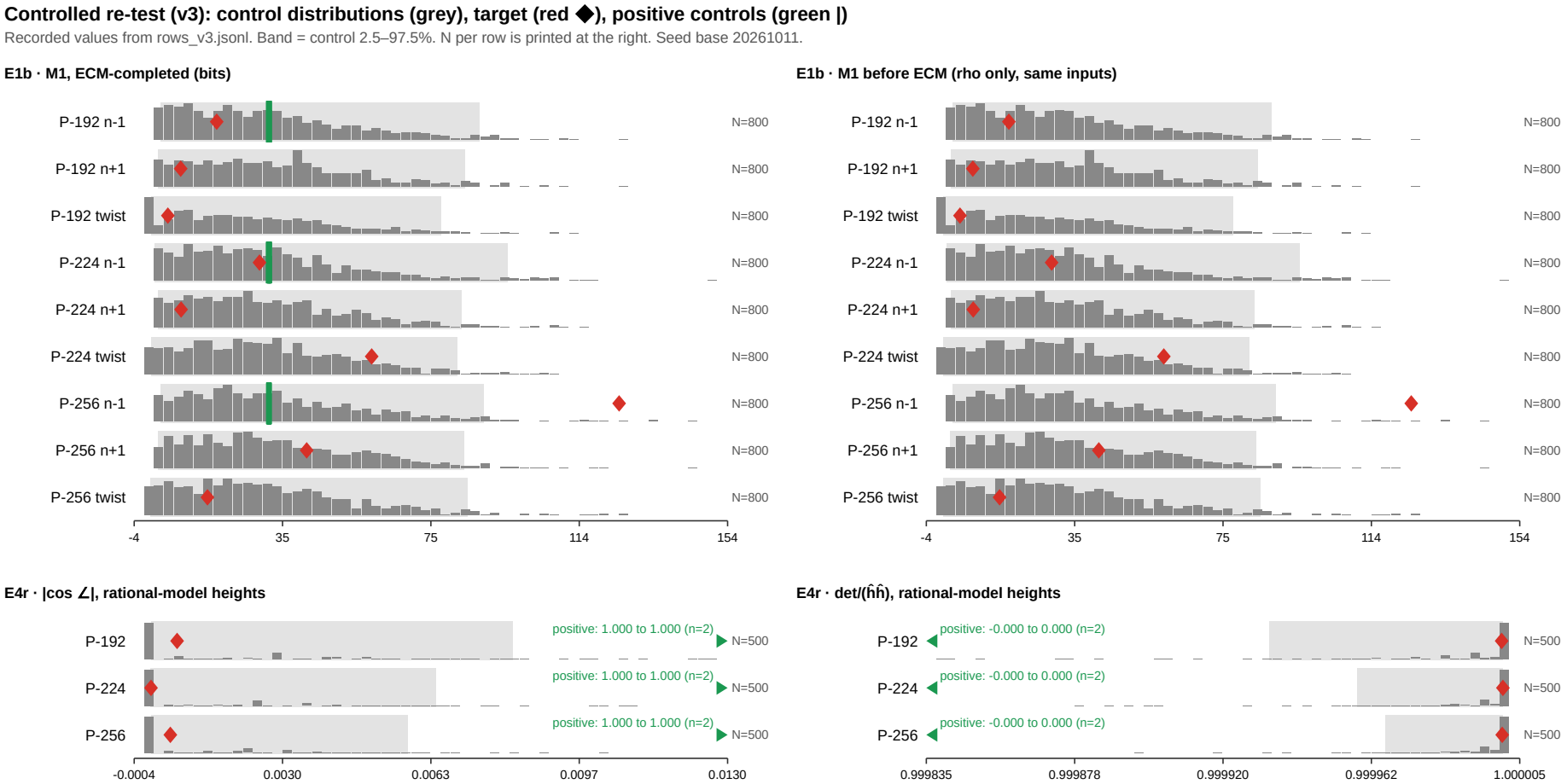
`AMENDMENT_2.md` was frozen (commit `6eb5c015a`, sha256 `6dba19da…1ee9`) after two dry runs and before any v3 run. Its disclosure section lists what the dry runs showed. The first full attempt was lost to a container restart after E4r P-192. The same binary was then re-run from scratch, and this rerun is the one reported. Wall time was 12 342 s, against a pre-flight estimate of about 3.6 h (E1b: 3919 s against about 70 min). Records: `rows_v3.jsonl`, `RESULTS_v3.md`, `results_v3.json`.

E1b, ECM-completed factoring. Rho with 2^18 iterations is followed by ECM (B1 = 2000, 20 curves) on every composite it leaves. The controls are the first 800 of v2's draws per curve.

- **Completeness.** ECM changed M1 for **0 of 2400** control values and for none of the targets. ECM could not split 1301, 1475 and 1592 composites per curve (P-192 / P-224 / P-256). These are large cofactors in which ECM at this budget found no prime below 2^32; the positive control shows it recovers such primes when they are present.
- **Positive control.** 150/150 planted 32-bit primes were recovered exactly.
- **Verdicts.** All 9 targets are within the control distribution at α = 0.05/9. P-256 `n − 1` ranks 796/800, p = 0.0125. So the v2 verdict does not depend on the factoring instrument's completeness.

E4r, pairing from rational-model heights.

- **Gate 1, dependent pairs: passed.** `(G̃, [2]G̃)` and `(G̃, [3]G̃)` give `|cos|` = 1 and det/(ĥĥ) = 0 to about 1e-16.
- **Gate 2, parallelogram law on 50 pairs per curve: passed.** The maximum defect is 1.35e-3 / 1.35e-3 / 2.71e-3 nats, and 45–47 of the 50 pairs are below 1e-10.
- **Target.** `|cos(G̃, R̃)|` is 5.9e-4 (P-192), 0 (P-224) and 4.4e-4 (P-256). Ranks are 316/500, 0/500 (ties at 0) and 313/500, with p = 0.74, 0.41 and 0.75: all within the control distribution.
- **The canonical pairing of random two-point lifts is exactly 0 for 59–66% of pairs** (296, 328 and 306 of 500). The nonzero values are small multiples of `log ℓ` for small primes ℓ, such as ½ ln 3, ½ ln 5, ½ ln 7, ½ ln 11 and (1/6) ln 2, and `|cos|` ≤ 0.013. This is a description of the recorded rows, not a tested claim.



Amendment 2 (v3): ECM-completed M1 and rational-model pairings

Figure (v3). The top two panels are E1b's M1 after and before ECM on the same inputs; they are identical. The bottom two are E4r's $|\cos|$ and det/(ĥĥ). Recorded values only.

What this does not show

Nothing here is a security statement. E1 is one incomplete small-factor metric against one null model of prime-order group orders. Its completeness below 2^32 is argued analytically: rho with 2^18 iterations per constant misses a 32-bit prime with probability of about $3 \cdot 10^{-4}$ per constant, and v3's ECM stage changed no control value. E2 is bounded by its census box. E3 and E4 compare against random points under one lift construction and say nothing about adversarially chosen lifts. "Within the control distribution" means the test did not distinguish the target from the null at the stated α. It does not mean the target is identical to the null.

Checklist (`controlled-experiment` §6)

- [x] every published sentence classified (`PROTOCOL.md` §0, table above)
- [x] comparative claims have a negative-control population with a stated null
- [x] every "nothing found" has a passing positive control on the same path (E2 90/90)
- [x] one instrument, identical budgets across arms
- [x] power check: v1 failed it (reported), v2 passes it
- [x] protocol frozen with hashes recorded (`results_v*.json`); dry run on scratch output
- [x] raw rows committed; summaries recomputable
- [x] decision rule and Bonferroni correction applied as written
- [x] fixed verdict wording; scope sentence present
- [x] superseded claims corrected in place in the note, dated 2026-10-11

Reproduce

```
cargo run --release --example nist_cyclotomic_controls -- --protocol v1   # ≈ 47 min on 4 vCPU
cargo run --release --example nist_cyclotomic_controls -- --protocol v2   # ≈ 2.5 h on 4 vCPU
cargo run --release --example nist_cyclotomic_controls -- --protocol v1 --plot
cargo run --release --example nist_cyclotomic_controls -- --protocol v2 --plot
cargo run --release --example nist_cyclotomic_controls -- --protocol v3       # ≈ 3.4 h on 4 vCPU
cargo run --release --example nist_cyclotomic_controls -- --protocol v3 --plot
sh research/nist_cyclotomic_controls_20261011/build.sh                    # REPORT.pdf
```

Controlled re-test: results (v3)

v3, `AMENDMENT_2.md`: E1b and E4r only. Seed base 20261011. Protocol file hashes are recorded in `results_v3.json`. Every verdict below is the decision rule applied mechanically.

E1b — M1 with ECM-completed factoring (rho 2^18, then ECM B1 = 2000 × 20 curves per composite)

Null: the first N = 800 of v2's Hasse-interval primes per curve (same seeds). Family: 9 targets × M1, Bonferroni α = 0.05/9 = 0.0056. Power check: min attainable p = 0.0025, passes.

curve	value	M1 target (rho → ECM)	control median [2.5%, 97.5%]	rank / N	p (two-sided)	verdict	controls whose M1 changed under ECM	unsplit composites (controls)
P-192	n − 1	17.5 → 17.5	27.8 [2.6, 87.7]	256/800	0.6417	within control distribution	0/800	457
P-192	n + 1	7.9 → 7.9	30.2 [2.0, 84.0]	100/800	0.2522	within control distribution	0/800	419
P-192	twist	4.5 → 4.5	24.4 [0.0, 77.5]	112/800	0.2896	within control distribution	0/800	425
P-224	n − 1	29.0 → 29.0	27.5 [1.0, 95.3]	420/800	0.9513	within control distribution	0/800	494
P-224	n + 1	8.1 → 8.1	27.6 [2.0, 83.1]	95/800	0.2397	within control distribution	0/800	493
P-224	twist	59.0 → 59.0	28.7 [0.0, 81.8]	712/800	0.2222	within control distribution	0/800	488
P-256	n − 1	125.0 → 125.0	28.7 [2.6, 88.8]	796/800	0.0125	within control distribution	0/800	526
P-256	n + 1	41.6 → 41.6	29.9 [2.0, 83.7]	543/800	0.6442	within control distribution	0/800	533
P-256	twist	15.1 → 15.1	27.6 [0.0, 84.6]	214/800	0.5368	within control distribution	0/800	533

Positive control (`q32 · q100 · r`, q32 a random prime in [2^31, 2^32)); pass = M1 equals log2(q32) exactly:

curve	detection
P-192	50/50 planted 32-bit primes recovered exactly
P-224	50/50 planted 32-bit primes recovered exactly
P-256	50/50 planted 32-bit primes recovered exactly

E4r — two-point-lift pairing from rational-model heights (k = 5; k = 4 for the accuracy check)

Null: the same N = 500 random pairs per curve as v2 (same seeds). Bonferroni α = 0.05/6 = 0.0083. Power check: min attainable p = 0.0040, passes. Accuracy gate (pre-registered): on the dependent pairs, abs(|cos| − 1) ≤ 1e-4 and abs(det/($\hat{h}\hat{h}$)) ≤ 1e-4 at k = 5 on every curve, else no orthogonality verdict.

curve	metric	target	random pairs median [2.5%, 97.5%] (N)	rank	p	verdict	dependent pairs ([2]G̃; [3]G̃)	median abs(k5 − k4) over controls

P-192: parallelogram defect over the first 50 control pairs: max abs = 1.35e-3 nats.

P-192 | |cos| | 5.888e-4 | 1.428e-16 [0.000e0, 8.157e-3] (500) | 316/500 | 0.739 | within control distribution | 1.000e0; 1.000e0 | 0.00e0 |
P-192 | det/(ĥĥ) | 1.000e0 | 1.000e0 [9.999e-1, 1.000e0] (500) | 184/500 | 0.739 | within control distribution | 0.000e0; -3.752e-16 | 0.00e0 |

P-224: parallelogram defect over the first 50 control pairs: max abs = 1.35e-3 nats.

P-224 | |cos| | 0.000e0 | 1.223e-16 [0.000e0, 6.437e-3] (500) | 0/500 | 0.407 | within control distribution | 1.000e0; 1.000e0 | 0.00e0 |
P-224 | det/(ĥĥ) | 1.000e0 | 1.000e0 [1.000e0, 1.000e0] (500) | 172/500 | 1.000 | within control distribution | 0.000e0; -5.521e-16 | 0.00e0 |

P-256: parallelogram defect over the first 50 control pairs: max abs = 2.71e-3 nats.

P-256 | |cos| | 4.359e-4 | 1.078e-16 [0.000e0, 5.788e-3] (500) | 313/500 | 0.750 | within control distribution | 1.000e0; 1.000e0 | 0.00e0 |
P-256 | det/(ĥĥ) | 1.000e0 | 1.000e0 [1.000e0, 1.000e0] (500) | 187/500 | 0.750 | within control distribution | 5.798e-16; -6.185e-16 | 0.00e0 |

Accuracy gate (dependent pairs): **passed**. Parallelogram gate (max abs defect ≤ 1e-2 nats on the first 50 control pairs per curve): **passed**.

Run time 12342 s.
Raw per-sample rows: `rows_v3.jsonl`.

Controlled re-test: results (v2)

v2, `PROTOCOL.md` as amended by `AMENDMENT_1.md`: larger N only. E2 is not re-run. Seed base 20261011. Protocol file hashes are recorded in `results_v2.json`. Every verdict below is the decision rule applied mechanically.

E1 — small-factor content (M1 = log2 of 2^32-smooth part found by factor(x, 2^18); M2 = # distinct such primes)

Null: N = 2000 random primes n' in the Hasse interval per curve. Bonferroni α = 0.05/18 = 0.0028. Power check: min attainable p = 2/(N+1) = 0.0010, passes.

curve	value	metric	target	control median [2.5%, 97.5%]	rank / N	p (two-sided)	verdict
P-192	n − 1	M1	17.5	29.4 [2.6, 87.2]	593/2000	0.5937	within control distribution
P-192	n − 1	M2	3.0	4.0 [1.0, 8.0]	343/2000	0.8056	within control distribution
P-192	n + 1	M1	7.9	29.9 [2.0, 84.0]	240/2000	0.2409	within control distribution

curve	value	metric	target	control median [2.5%, 97.5%]	rank / N	p (two-sided)	verdict
P-192	n + 1	M2	3.0	4.0 [1.0, 8.0]	303/2000	0.7326	within control distribution
P-192	twist	M1	4.5	25.4 [0.0, 76.1]	292/2000	0.3058	within control distribution
P-192	twist	M2	1.0	2.0 [0.0, 5.0]	206/2000	0.6557	within control distribution
P-224	n − 1	M1	29.0	29.1 [2.0, 88.8]	991/2000	0.9915	within control distribution
P-224	n − 1	M2	5.0	4.0 [1.0, 8.0]	1278/2000	0.7226	within control distribution
P-224	n + 1	M1	8.1	28.9 [2.0, 85.7]	238/2000	0.2399	within control distribution
P-224	n + 1	M2	3.0	4.0 [1.0, 8.0]	337/2000	0.7816	within control distribution
P-224	twist	M1	59.0	27.7 [0.0, 83.1]	1791/2000	0.2099	within control distribution
P-224	twist	M2	5.0	3.0 [0.0, 7.0]	1623/2000	0.3778	within control distribution
P-256	n − 1	M1	125.0	28.7 [2.6, 90.2]	1992/2000	0.0090	within control distribution
P-256	n − 1	M2	10.0	4.0 [1.0, 8.0]	1991/2000	0.0100	within control distribution
P-256	n + 1	M1	41.6	28.9 [2.0, 82.8]	1387/2000	0.6137	within control distribution
P-256	n + 1	M2	4.0	4.0 [1.0, 8.0]	791/2000	1.0000	within control distribution
P-256	twist	M1	15.1	27.3 [0.0, 85.0]	566/2000	0.5667	within control distribution
P-256	twist	M2	4.0	3.0 [0.0, 7.0]	1308/2000	0.6927	within control distribution

Positive control (planted 2^20-smooth part of ≈120 bits):

curve	detection
P-192	20/20 planted-smooth values above the control 97.5th percentile of M1 (87.2)
P-224	20/20 planted-smooth values above the control 97.5th percentile of M1 (88.8)
P-256	20/20 planted-smooth values above the control 97.5th percentile of M1 (90.2)

E3 — $\hat{h}$ of the one-point lift: generator vs random points (nats; $\hat{h} := \lim h_x([2^k]P)/4^k$)

Bonferroni α = 0.05/6 = 0.0083. Power check: min attainable p = 2/(N+1) = 0.0020, passes.

curve	metric	$\tilde{G}$	random points median [2.5%, 97.5%] (N)	rank	p	verdict	small-x positive control median [min, max]
P-192	$\hat{h}$/ln p	0.9865	0.9923 [0.9662, 1.0034] (1000)	281/1000	0.563	within control distribution	0.6600 [0.6463, 0.6653]
P-192	h_x − $\hat{h}$	-0.5493	0.1438 [-0.5493, 2.1972] (1000)	43/1000	0.438	within control distribution	-65.9053 [-69.2966, -65.0771]
P-224	$\hat{h}$/ln p	0.9858	0.9938 [0.9729, 1.0031] (1000)	188/1000	0.378	within control distribution	0.6615 [0.6471, 0.6681]

curve	metric	$\tilde{G}$	random points median [2.5%, 97.5%] (N)	rank	p	verdict	small-x positive control median [min, max]
P-224	$h_x - \hat{h}$	1.8637	0.1438 [-0.5493, 2.2034] (1000)	956/1000	0.090	within control distribution	-80.8559 [-83.0641, -78.5936]
P-256	$\hat{h}/\ln p$	0.9982	0.9941 [0.9746, 1.0025] (1000)	737/1000	0.527	within control distribution	0.6625 [0.6444, 0.6684]
P-256	$h_x - \hat{h}$	-0.5493	0.1438 [-0.5493, 2.1715] (1000)	0/1000	0.110	within control distribution	-96.1205 [-101.4126, -92.3954]

E4 — two-point-lift pairing: ($\tilde{G}$, $\tilde{R}$) vs random pairs

Bonferroni $\alpha = 0.05/6 = 0.0083$. Power check: min attainable $p = 2/(N+1) = 0.0040$, passes.

curve	metric	target	random pairs median [2.5%, 97.5%] (N)	rank	p	verdict	dependent-pair positive control ([2]$\tilde{G}$; [3]$\tilde{G}$)
P-192		cos		0.00071	0.00130 [0.00071, 0.00904] (500)	14/500	0.060
P-192	$\det(\hat{h}\hat{h})$	1.00000	1.00000 [0.99992, 1.00000] (500)	486/500	0.060	within control distribution	-4.93e-3; -2.15e-3
P-224		cos		0.00129	0.00130 [0.00071, 0.00740] (500)	132/500	0.531
P-224	$\det(\hat{h}\hat{h})$	1.00000	1.00000 [0.99995, 1.00000] (500)	368/500	0.531	within control distribution	-4.92e-3; -2.14e-3
P-256		cos		0.00173	0.00130 [0.00051, 0.00592] (500)	363/500	0.551
P-256	$\det(\hat{h}\hat{h})$	1.00000	1.00000 [0.99996, 1.00000] (500)	137/500	0.551	within control distribution	-4.90e-3; -2.16e-3

Run time 8966 s. Raw per-sample rows: `rows_v2.jsonl`.

Controlled re-test: results (v1)

v1, `PROTOCOL.md` exactly as frozen. Its N fail the power check, so no rank test here can reach its α (see the power column). Seed base 20261011. Protocol file hashes are recorded in `results_v1.json`. Every verdict below is the decision rule applied mechanically.

E1 — small-factor content (M1 = log2 of 2^32-smooth part found by factor(x, 2^18); M2 = # distinct such primes)

Null: N = 400 random primes n’ in the Hasse interval per curve. Bonferroni α = 0.05/18 = 0.0028. Power check: min attainable p = 2/(N+1) = 0.0050, **FAILS (no target can be declared unusual)**.

curve	value	metric	target	control median [2.5%, 97.5%]	rank / N	p (two-sided)	verdict
P-192	n − 1	M1	17.5	29.1 [2.6, 89.9]	117/400	0.5885	within control distribution
P-192	n − 1	M2	3.0	4.0 [2.0, 8.0]	70/400	0.8429	within control distribution
P-192	n + 1	M1	7.9	30.6 [2.0, 83.9]	46/400	0.2344	within control distribution
P-192	n + 1	M2	3.0	4.0 [1.0, 7.0]	61/400	0.7282	within control distribution
P-192	twist	M1	4.5	23.9 [0.0, 74.9]	52/400	0.2793	within control distribution
P-192	twist	M2	1.0	2.0 [0.0, 5.0]	39/400	0.6983	within control distribution
P-224	n − 1	M1	29.0	27.9 [2.0, 89.7]	206/400	0.9726	within control distribution
P-224	n − 1	M2	5.0	4.0 [1.0, 7.0]	262/400	0.6933	within control distribution
P-224	n + 1	M1	8.1	28.7 [2.6, 78.0]	40/400	0.2045	within control distribution
P-224	n + 1	M2	3.0	4.0 [2.0, 8.0]	60/400	0.7581	within control distribution
P-224	twist	M1	59.0	26.7 [0.0, 76.4]	360/400	0.2045	within control distribution
P-224	twist	M2	5.0	3.0 [0.0, 6.0]	336/400	0.3242	within control distribution
P-256	n − 1	M1	125.0	28.7 [2.6, 84.3]	399/400	0.0100	within control distribution
P-256	n − 1	M2	10.0	4.0 [2.0, 8.0]	399/400	0.0100	within control distribution
P-256	n + 1	M1	41.6	31.4 [3.6, 88.2]	258/400	0.7132	within control distribution
P-256	n + 1	M2	4.0	4.0 [1.0, 8.0]	149/400	1.0000	within control distribution
P-256	twist	M1	15.1	27.9 [0.0, 84.3]	118/400	0.5935	within control distribution
P-256	twist	M2	4.0	3.0 [0.0, 7.0]	246/400	0.7731	within control distribution

Positive control (planted 2^20-smooth part of ≈120 bits):

curve	detection
P-192	20/20 planted-smooth values above the control 97.5th percentile of M1 (89.9)
P-224	20/20 planted-smooth values above the control 97.5th percentile of M1 (89.7)
P-256	20/20 planted-smooth values above the control 97.5th percentile of M1 (84.3)

E2 — census power (|x| ≤ 10^6; x = u/d², 2 ≤ d ≤ 40, |u| ≤ 10^5)

curve	planted integer found	planted d=2 found	planted d=3 found	random-b curves with any find	NIST b finds
P-192	10/10	10/10	10/10	0/10	0
P-224	10/10	10/10	10/10	0/10	0
P-256	10/10	10/10	10/10	0/10	0

E3 — ĥ of the one-point lift: generator vs random points (nats; ĥ := lim h_x([2^k]P)/4^k)

Bonferroni α = 0.05/6 = 0.0083. Power check: min attainable p = 2/(N+1) = 0.0100, **FAILS (no target can be declared unusual)**.

curve	metric	G̃	random points median [2.5%, 97.5%] (N)	rank	p	verdict	small-x positive control median [min, max]
P-192	ĥ/ln p	0.9865	0.9922 [0.9616, 1.0023] (200)	59/200	0.597	within control distribution	0.6613 [0.6438, 0.6694]
P-192	h_x − ĥ	-0.5493	0.1438 [-0.5493, 2.4836] (200)	11/200	0.498	within control distribution	-66.5123 [-68.9485, -64.2055]
P-224	ĥ/ln p	0.9858	0.9943 [0.9751, 1.0032] (200)	40/200	0.408	within control distribution	0.6611 [0.6453, 0.6691]
P-224	h_x − ĥ	1.8637	0.1438 [-0.5493, 2.0999] (200)	191/200	0.100	within control distribution	-81.0740 [-83.1827, -79.1396]
P-256	ĥ/ln p	0.9982	0.9947 [0.9733, 1.0024] (200)	145/200	0.557	within control distribution	0.6625 [0.6508, 0.6683]
P-256	h_x − ĥ	-0.5493	0.0000 [-0.5493, 2.8883] (200)	0/200	0.129	within control distribution	-96.0648 [-99.8508, -93.5027]

E4 — two-point-lift pairing: (G̃, R̃) vs random pairs

Bonferroni α = 0.05/6 = 0.0083. Power check: min attainable p = 2/(N+1) = 0.0198, **FAILS (no target can be declared unusual)**.

curve	metric	target	random pairs median [2.5%, 97.5%] (N)	rank	p	verdict	dependent-pair positive control ([2]G̃; [3]G̃)
P-192		cos		0.00071	0.00130 [0.00128, 0.00688] (100)	0/100	0.020
P-192	det/(ĥĥ)	1.00000	1.00000 [0.99995, 1.00000] (100)	100/100	0.020	within control distribution	-4.93e-3; -2.15e-3
P-224		cos		0.00129	0.00130 [0.00070, 0.01129] (100)	22/100	0.455
P-224	det/(ĥĥ)	1.00000	1.00000 [0.99987, 1.00000] (100)	78/100	0.455	within control distribution	-4.92e-3; -2.14e-3
P-256		cos		0.00173	0.00130 [0.00046, 0.00795] (100)	69/100	0.634
P-256	det/(ĥĥ)	1.00000	1.00000 [0.99994, 1.00000] (100)	31/100	0.634	within control distribution	-4.90e-3; -2.16e-3

Run time 2804 s.
Raw per-sample

rows: `rows_v1.jsonl`.

Appendix A: PROTOCOL.md (frozen)

Protocol: controlled re-test of the comparative claims in RESEARCH_NIST_CYCLOTOMIC_POINT_HEIGHTS.md

Status: frozen before any data was collected (2026-10-11). Follows `.claude/skills/controlled-experiment/SKILL.md`. Any change after the first run is a new protocol version, reported alongside this one.

0. Claim classification of the original note

original claim (note §)	class	action here
`f ± 1` factor in `Z[t]` with `Φ_1, Φ_2, Φ_4` (§1)	identity (certified)	none
full factorisations of `p ± 1`, `n ± 1`, twist, `t`, `4p − t²` (§1, §2)	exact computation	none
embedding degrees `(n−1)/10, (n−1)/3, (n−1)/3` (§2.1)	exact computation	none
Cheon cost tables (§2.2)	exact computation given the factorisation	none
“P-256’s `n − 1` is an outlier / 1-in-30 accident” (§0, §2.2)	**comparative**	**E1**
“`N_d`, twist: nothing small enough to matter” (§2.3)	**comparative** (twist part)	**E1** (twist)
“canonical lift has no small points” (§3.1)	**“nothing found” without positive control**	**E2**
“lifted generator $\hat{h} \approx \log p$; nowhere near small” (§3.2)	**comparative**	**E3**
“two-point lift independent / orthogonal” (§3.3)	**comparative**	**E4**
“the cyclotomic structure never reaches the points” (§0)	out of scope as stated	reword to the exact statement (MOV needs k; k is exact)

E1. Small-factor content of `n − 1`, `n + 1` and the twist order

- **Targets.** P-192, P-224, P-256: `n − 1`, `n + 1`, `T = 2p + 2 − n` (9 values).
- **Null.** For each curve’s `p`, group orders of prime-order curves over `F_p` are modelled as **uniform random primes `n'` in the Hasse interval `[p + 1 − 2√p, p + 1 + 2√p]`**. Control values are `n' − 1`, `n' + 1`, `2p + 2 − n'`, computed identically. `N = 400` controls per curve. Seed `20261011 + curve index`.
- **Instrument (identical on every arm).** `curve_traits::arith::factor(x, 2^18)`: trial division to 2^16 then Brent rho, 2^18 iterations per composite.
- **Metric M1.** `log2` of the product, with multiplicity, of the returned prime factors that are `< 2^32`. M1 is defined on what the instrument returns. It is incomplete, but the incompleteness is identical across arms.
- **Metric M2.** The number of distinct returned prime factors `< 2^32`.
- **Positive control.** For each curve, 20 constructed values `x = s · q`. Here `s` is a random 2^20-smooth product of about 120 bits, and `q` is a random prime that brings `x` to the size of `n`. They must score M1 above the 97.5th percentile of the controls. Expected: yes.

- **Decision rule.** The empirical upper-tail fraction `(1 + #{control ≥ target}) / (N + 1)` and the matching lower tail. Two-sided `p = 2·min(tails)`. Bonferroni over 9 targets × 2 metrics = 18 tests, α = 0.05 / 18. A target is **unusual** only if `p < α`. Otherwise it is **within the control distribution (rank r/N)**.

E2. Small-point census power on the canonical lift

- **Targets.** $y^2 = x^3 - 3x + b$ with the NIST b, for all three curves.
- **Instrument.** The census in `examples/nist_cyclotomic_point_heights.rs`, unchanged. It scans integer $|x| \le 10^6$, and rational $x = u/d^2$ with $2 \le d \le 40$ and $|u| \le 10^5$, behind the residue filter.
- **Positive control.** For each curve's p, plant points on 30 curves with b of the same size as the NIST b. 10 curves get an integer point (x_0 uniform in $|x_0| \le 10^6$, y_0 uniform below $\sqrt{p}$, $b = y_0^2 - x_0^3 + 3x_0$). 10 get a rational point with $d = 2$ and 10 with $d = 3$ (u uniform in $|u| \le 10^5$ with $\gcd(u, d) = 1$, v near $\sqrt{(p\ d^6)}$, $b = (v^2 - u^3 + 3u\ d^4)/d^6$ when divisible). **Pass:** every planted point is found.
- **Negative control.** 30 curves with b uniform in $[0, p)$ and no planted point. Expect zero finds.
- **Verdict.** If the positive control passes, "no point in this census" is reported for the NIST curves with the census bounds stated. It never becomes "no small point".

E3. Canonical height of the lifted generator vs lifted random points

- **Target.** $\hat{h}(\tilde{G})/\ln p$ on the one-point lift $y^2 = x^3 - 3x + B$ through $(G_x, G_y) \in Z^2$, using the same height routine as the note (5 doublings).
- **Null.** $N = 200$ uniform random points $R \in E(F_p)$ per curve. Each is lifted identically through (x_R, y_R) with representatives in $[0, p)$. Seed fixed.
- **Positive control.** 20 points per curve with $x_R < 2^{32}$ (x uniform in $[1, 2^{32})$ until it lands on the curve), lifted identically. Expected: h_x is much smaller. $\hat{h}$ is reported as found.
- **Metrics.** $\hat{h}/\ln p$ and $h_x - \hat{h}$.
- **Decision rule.** Rank of the target in the null, two-sided, Bonferroni over 3 curves × 2 metrics.

E4. Height pairing of two-point lifts

- **Target.** $|\cos \angle(\tilde{G}, \tilde{R})|$ and the normalised regulator $\det / (\hat{h}(\tilde{G})\hat{h}(\tilde{R}))$ on the two-point lift through $\tilde{G}$ and $\tilde{R}$ ($R = [0x1234567890abcdef]G$).
- **Null.** $N = 100$ two-point lifts through pairs of uniform random points per curve, constructed identically.
- **Positive control.** On the one-point lift E_1 through $\tilde{G}$, the genuinely dependent rational pairs $(\tilde{G}, [2]\tilde{G})$ and $(\tilde{G}, [3]\tilde{G})$. Expected: $|\cos| = 1$ and normalised regulator 0 to numerical precision.
- **Decision rule.** Rank in the null, two-sided, Bonferroni over 3 × 2.

What this protocol cannot show

Nothing here is a security statement. E1 measures one incomplete small-factor metric. E2 is bounded by its census box. E3 and E4 compare against random points under one lift construction, and say nothing about lifts chosen adversarially.

Appendix B: AMENDMENT_1.md (frozen)

Amendment 1 to PROTOCOL.md: sample sizes that can reject

Status: frozen 2026-10-11, before any run at protocol N, v1 or v2. `PROTOCOL.md` is not edited. Both versions are run and reported:

- **v1** is `PROTOCOL.md` as frozen.
- **v2** is `PROTOCOL.md` with only the changes below.

Why

A dry run at small N (`--protocol quick`, scratch output, not committed) showed that v1's decision rule cannot reject in any comparative family. With the add-one empirical tail the smallest attainable two-sided p is `2/(N + 1)`:

family	tests m	α/m	v1 N	v1 min p	v2 N	v2 min p
E1	18	0.00278	400	0.00499 (**cannot reject**)	2000	0.00100
E3	6	0.00833	200	0.00995 (**cannot reject**)	1000	0.00200
E4	6	0.00833	100	0.01980 (**cannot reject**)	500	0.00399

v1 can therefore only ever report "within the control distribution", and its nulls are vacuous. That is a design error in the protocol, not a result. v2's N are set by the rule now in the skill, `N + 1 ≥ 4m/α`: 1440, 480 and 480, rounded up. No result entered the choice.

What changes in v2

- E1: `N = 2000` controls per curve (was 400).
- E3: `N = 1000` random points per curve (was 200).
- E4: `N = 500` random pairs per curve (was 100).
- E2 is not re-run. It has no rank test, and v1's E2 stands as v2's.

Nothing else changes: instruments, budgets, metrics, positive controls, seeds (`20261011 + offset`), α, the Bonferroni families and the verdict wording are all as in `PROTOCOL.md`. Because every control sample is drawn first from its seeded stream, v1's control samples are exact prefixes of v2's for E1, E3 and E4. The positive-control draws come after the controls and differ between versions.

Disclosure: what the dry run showed before this amendment

The dry run used N = 40 / 20 / 10 for E1 / E3 / E4, 5 planted values per curve for the E1 and E3 positive controls, and 3 curves per kind for E2. Its output was looked at. In particular:

- **E1.** Target M1 values: P-256 `n − 1` 125.0, against a dry-run control maximum of 91.1 (N = 40). All other targets were inside the dry-run control range. Planted positive controls scored M1 128–137.
- **E2.** Every planted point was found (27/27). No random-`b` curve gave a find (0/9).
- **E3.** Small-`x` positive controls had `ĥ/ln p` 0.65–0.67 and `h_x − ĥ` from −65 to −97 nats, against a generator `ĥ/ln p` of 0.99–1.00.
- **E4.** P-192 and P-224 dry-run rows were produced. Their summary statistics were not examined before this amendment.

None of this changes v2's N, which the power rule fixes. It is recorded so that a reader can discount v2 for having been designed after seeing it.

Implementation notes (clarifications, not changes)

- E2 planted rational points draw `v` uniformly in `[0, √(p·d⁶))`. The protocol's "near `√(p d⁶)`" means the same bit size range, so `b` is at most `p` in absolute value, as for the NIST `b`.
- Instrument calls run in parallel across threads. Samples are drawn sequentially from the seeded stream first, and results are collected in sample order, so the output does not depend on the thread count.

Appendix C: AMENDMENT_2.md (frozen)

Amendment 2: an ECM-completed E1 instrument (E1b) and an unbiased E4 height (E4r)

Status: frozen 2026-10-12, before any v3 run. It is dated after two dry runs on scratch output (disclosed below). `PROTOCOL.md` and `AMENDMENT_1.md` are not edited. This amendment adds two experiments. It reports them as **v3** alongside v1 and v2 and does not replace them.

Why

- **E1b.** E1's M1 is defined on what budgeted rho returns. A reader can fairly ask whether P-256 `n − 1`'s rank (1992/2000 in v2) is an artefact of that incompleteness. Analytically, Brent rho with 2^18 iterations per constant misses a given prime below 2^32 with probability of about `exp(−2^36/2^33) ≈ 3·10⁻⁴` per constant, so M1 should already be essentially complete. E1b tests that with a second factoring stage.
- **E4r.** v1/v2's E4 instrument has a floor. Random pairs pile up at `|cos|` = 0.0013, and exactly dependent pairs read 1.0011–1.0025. `height_sequence` takes the naive height on the denominator-cleared model `(u²x, u³y)`, where `u = lcm` of all denominators including the point's own. That inflates `h_x` by about `2 log u`, and 4⁵ doublings do not remove it. E4r takes the naive height of the reduced rational `x` on the original model. The canonical limit is unchanged, the bias is removed, and the cost is the same.

E1b

- **Targets.** The 9 values of E1 (`n − 1`, `n + 1` and twist order for three curves).
- **Null.** The first **N = 800** Hasse-interval primes of v2's E1 stream (seed `20261011 + curve index`, drawn sequentially), so the same inputs as v2's first 800 controls.

- **Instrument (identical on every arm).** `arith::factor(x, 2^18)` as in E1. Then every composite it leaves gets ECM with `B1 = 2000` and 20 curves (`cryptanalysis::ecm::ecm_factor`, seeds fixed per value), recursively on the pieces until each is prime or ECM fails.
- **Metric.** M1 as in E1, on the ECM-completed factor list. Rows also record the rho-only M1 on the same input, so completeness is measured directly. M2 is recorded but not tested.
- **Positive control.** 50 values per curve of the form `q32 · q100 · r`. `q32` is a random prime in `[2^31, 2^32)`, `q100` a random 100-bit prime and `r` a random prime bringing the value to the size of `n` (seed `20261011 + 400 + curve index`). **Pass:** M1 equals `log2 q32` exactly on at least 49/50 per curve.
- **Decision rule.** Two-sided add-one rank as in E1. The family is 9 targets × M1, so α = 0.05/9 = 0.0056. Power check: `2/(N + 1)` = 0.0025 < 0.0056, and `N + 1 ≥ 4m/α = 720` holds. Verdict wording as in `PROTOCOL.md`.
- **What counts as "robust".** P-256 `n − 1`'s v2 verdict ("within the control distribution") is robust to instrument completeness if E1b reaches the same verdict, and if fewer than 1% of control values change M1 under ECM.

E4r

- **Targets, null, seeds, positive controls.** As in v2's E4: the same 500 random pairs per curve (seed `20261011 + 300 + curve index`), the target `(G̃, R̃)` with `R = [0x1234567890abcdef]G`, and the dependent pairs `(G̃, [2]G̃)` and `(G̃, [3]G̃)` on `E_1`.
- **Instrument.** `height_sequence_on(…, 5, rational = true)`: the same Jacobian doubling, with `h_x` taken of `X/(Z²u²)` in lowest terms. The metrics are `|cos|` and det/(ĥĥ) at k = 5, as before. The k = 4 values are recorded for the accuracy check.
- **Accuracy gates (pre-registered; both must pass, else no orthogonality verdict).**
  1. **Dependent pairs.** On every curve, abs(`|cos|` − 1) ≤ 1e-4 and abs(det/(ĥĥ)) ≤ 1e-4 at k = 5.
  2. **Parallelogram law.** On the first 50 control pairs per curve, `max |ĥ(P+Q) + ĥ(P−Q) − 2ĥ(P) − 2ĥ(Q)|` ≤ 1e-2 nats.
- **Decision rule.** Rank as in E4. The family is 3 curves × 2 metrics, so α = 0.05/6. N = 500 passes the power check (`N + 1 ≥ 480`).
- **What this licenses.** If both gates pass, the measured `|cos|` of `(G̃, R̃)` and its rank among random pairs replace the withdrawn "cos < 0.002" sentence. If either gate fails, the note keeps "orthogonality unresolved".

Pre-flight runtime estimate (from the dry runs, 4 vCPU)

family	per-sample cost	fixed cost	estimate at protocol N
E1b	≈ 2.3 s CPU per value (one ECM pass of 20 curves on the composite rho leaves)	negligible	3 × (2400 + 3 + 50) × 2.3 s / 4 ≈ 70 min
E4r	≈ 13 / 18 / 23 s CPU per pair (P-192 / P-224 / P-256); parallelogram adds a third for 50 pairs per curve	dependent-pair heights (`[3]G̃`, `[4]G̃`), single-threaded, ≈ 5 / 7 / 9 min	≈ 2.5 h
total			≈ 3.6 h

The actual run time will be recorded next to this estimate.

Disclosure: what the dry runs showed before this amendment

Both dry runs used N = 40 (E1b) and N = 10 (E4r) with 5 E1b positives per curve.

- **E1b.**
  - No control value's M1 changed under ECM (0/120 per curve).
  - The targets' M1 were unchanged: P-256 `n − 1` stays 125.0.
  - P-256 `n − 1` ranked 40/40 among the 40 dry-run controls.
  - 15/15 planted 32-bit primes were recovered.
  - 20–27 composites per curve stayed unsplit after ECM. Those are large cofactors, irrelevant to M1.
- **E4r.**
  - Dependent pairs gave `|cos|` = 1 and det = 0 to about 1e-16, and the parallelogram law held to about 1e-13 on most pairs.
  - The canonical pairings of random two-point lifts came out as **exact small multiples of `log ℓ` or exactly 0**: ½ ln 3, ½ ln 5, ½ ln 7, ½ ln 11, and about (1/6) ln 2 with a residual of 1.4e-3 nats.
  - Target `|cos|`: 5.9e-4 (P-192), 0 (P-224), 4.4e-4 (P-256).
  - So `|cos|` is a discrete-valued statistic with many ties. The rank rule handles ties conservatively.

None of this changes the design above, which v2 fixed except for the instrument. It is recorded so readers can discount v3 for having been specified after seeing it.